

Rohan

Bengaluru, India

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EDUCATION

PES University

Bachelor of Technology in Computer Science & Engineering

Bengaluru, India

2023 – 2027

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Compiler Design, Cloud Computing, Generative AI, Object-Oriented Analysis and Design, Software Engineering.

EXPERIENCE

Research Intern

May 2025 – Present

CoDMAV Research Lab, PES University

Bengaluru, India

- Engineered an **AMLSim** replay pipeline with controllable fraud burst injection, enabling fully reproducible experimentation across all evaluation scenarios for an anti-money laundering detection system.
- Built a **FastAPI** async serving layer with streaming cache management, migrating from in-memory Python dict to **Redis** for scalable, low-latency inference throughput.
- Implemented all **5 evaluation scenarios** and baseline systems end-to-end, forming the empirical backbone of an active research publication. Contributing to paper sections on experimental setup, baseline comparison, and ablation results. (*manuscript in preparation*)

PROJECTS

Policy-Aware Code Compliance Auditor | [GitHub](#) | *PyTorch, CodeBERT, LLaMA 3.1, HuggingFace, FAISS, RAG* Apr 2026

- Built a hybrid **RAG + classifier** system that auto-audits document databases against business policies, achieving a **93% compliance detection rate** across live testing with scalable batch inference.
- Fine-tuned **CodeBERT** with **LoRA (PEFT)** as a fast-gating routing engine, reducing **LLM API calls by 60%** through intelligent request filtering and batched inference optimization.
- Reduced inference costs by **75%** via **Gemini Embeddings + FAISS** semantic retrieval, ensuring only the most relevant policy context reached the LLM and eliminating redundant API calls.

Airline Reservation System | [GitHub](#) | *React, Spring Boot, Java 17, Firebase Firestore, Docker*

Apr 2026

- Built a full-stack booking platform with a **React/Vite** frontend and **Spring Boot** REST backend; containerized with **Docker**, reducing environment setup time from hours to minutes.
- Engineered a real-time interactive seat map resolving concurrent bookings with **zero conflicts** using **Firebase Firestore** transactions, applying **GRASP** design patterns for clean separation of concerns.

Real-Time Streaming Platform | [GitHub](#) | *MERN Stack, WebSockets, Node.js*

Jul 2025

- Architected a live streaming app achieving **sub-100ms latency** using **WebSockets** and **Node.js**, handling multiple concurrent text and video streams with secure private room isolation and **zero data leakage**.
- Built fault-tolerant reconnection logic to recover stream failures gracefully under unstable network conditions, ensuring high availability across concurrent sessions.

Smart Glove for Assistive Healthcare Communication | [GitHub](#) | *C++, Arduino, I2C, Bluetooth*

Jan 2025

- Built a wearable device converting sign language to real-time text using an Arduino, 4 flex sensors, and MPU6050 via **I2C**; delivered wireless output via **HC-05 Bluetooth** with zero external dependencies, enabling hearing-impaired patients to communicate independently in clinical settings.
- Trained a gesture-classification model on sensor data to map flex and orientation readings to **25+ hand signs** with **90%+** recognition accuracy, optimizing sampling rate and calibration for real-time on-device inference.

LEADERSHIP & EXTRACURRICULAR

Vice Chair & Secretary

August 2025 – June 2026

IEEE PESU Student Branch

Bengaluru, India

- Led technical direction for **200+** students, designing an advanced AI and cybersecurity curriculum delivered across workshops and hands-on sessions; managed all CTF competition infrastructure end-to-end.
- Mentored junior members via structured code reviews, helping **3+** members ship their first open-source projects and accelerating team proficiency in ML and software engineering.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript (ES6+), SQL

AI / ML: PyTorch, HuggingFace Transformers, CodeBERT, LLaMA 3.1, LoRA/PEFT, FAISS, RAG, NLP Pipelines, LLM Fine-Tuning, Transformer Architectures, Scikit-learn, NumPy, Pandas, Prompt Engineering

Web & Backend: React.js, Node.js, Express.js, Spring Boot, FastAPI, REST APIs, WebSockets, Docker, Redis, MongoDB, Firebase Firestore, HTML5, CSS3

Tools: Git, GitHub, VS Code, Postman, Figma, Linux/Bash, Vercel, Docker